A Fast Algorithm for Generating All Triangulations of Biconnected Labeled Plane Graphs

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Overview

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Introduction

The Problem:

- Generate **all possible triangulations** of biconnected plane graphs
- Previous methods fail with complex structures due to multi-edge creation
- Important applications in graphics, engineering, and chip design

Our Contribution:

- Novel algorithm with **constant time per triangulation**
- Solves the multi-edge problem using **safe root** concept
- Practical for real-world applications

Algorithm

The complete algorithm will need us different methods

- Cycle Graphs
- Biconnected Graphs
- find all the triangulations of a biconnected graph
- find a safe root for a biconnected graph
- find all the triangulations of a biconnected graph using safe root
- find all the triangulations of a biconnected graph using safe root in constant time

Find All triangulations

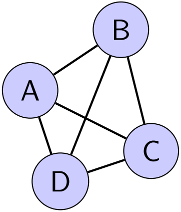
find all child triangulations of the root
for each child triangulation
$=0$

Cycle Graphs

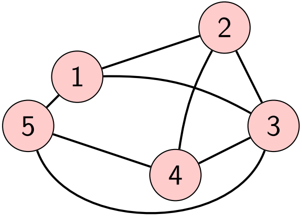
1: find all child triangulations of the root $=0$

Example Graphs

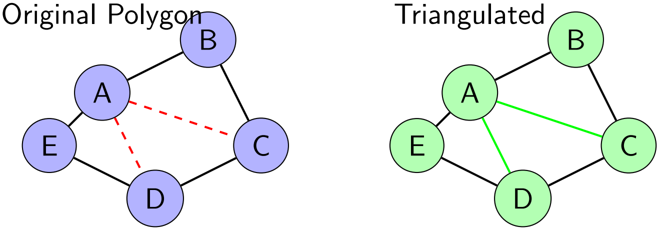
Simple Biconnected Graph



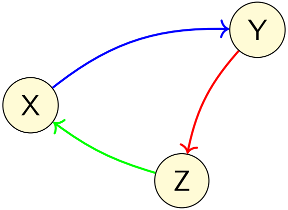
Complex Graph with Curved Edges



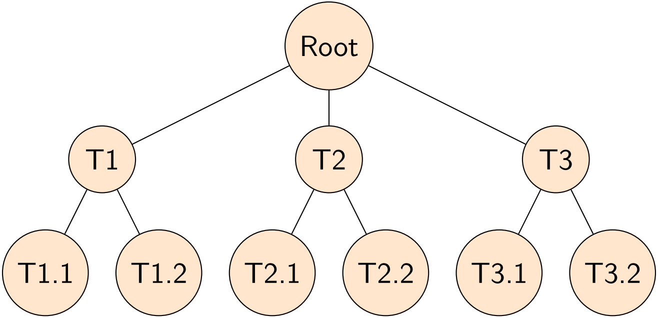
Step-by-step triangulation example:



Non-straight edge example:



Tree structure of triangulations:



Graph with curved edges: